

CASE STUDY: LGU DOCUMENT MANAGEMENT SYSTEM

System objectives

1. A database in which Sangguniang Bayan employees can store and arrange digital copies of constitutional documents.
2. Implement an advanced search function that enables users to efficiently locate specific documents based on keywords or categories.
3. Develop a module named Ordinances, a module named Resolutions, a module named Committee Reports, a module named Privilege Speeches, a module named Journals, a module named Internal Rules, and another module named Codes that contain files and images in their viewpoint labels, all of the modules listed above will facilitate and manage the filling process by providing a function to view, print, edit, update, and delete it from the system. Provide storage from the system and one drive that serves as a backup and restores all uploaded documents.

System functions

Admin. Serves as the person in charge of the system's inner workings, from approving new users to changing system files. The administrator can also remove accounts from the system.

User. The people who will use the system on a daily basis; can create an account and have it approved by the administrator, and can only utilize the system's key features such as viewing.

Log-in Security. This Feature serves as a security measure to ensure that all users of this system are verified employees of the Sangguniang Bayan and office of the Vice-Mayor.

View. The Admin and the User account both have a function to view documents and images in each module.

Search. The search function allows the user to be able to locate the file or document easily by just inputting key-words or resolution and ordinance numbers.

Search in open documents. The search function allows the user to be able to locate key-words in the opened document.

Delete. If the user is the system administrator, the admin can delete files and images from any module as well as other modules such as the accounts Module by deleting any type of accounts.

Update. If the user is the administrator, the admin can make changes documents and images and to the Account module by updating username or password of any type of account.

Upload. If the user is the administrator, the admin can upload documents and images to the system.

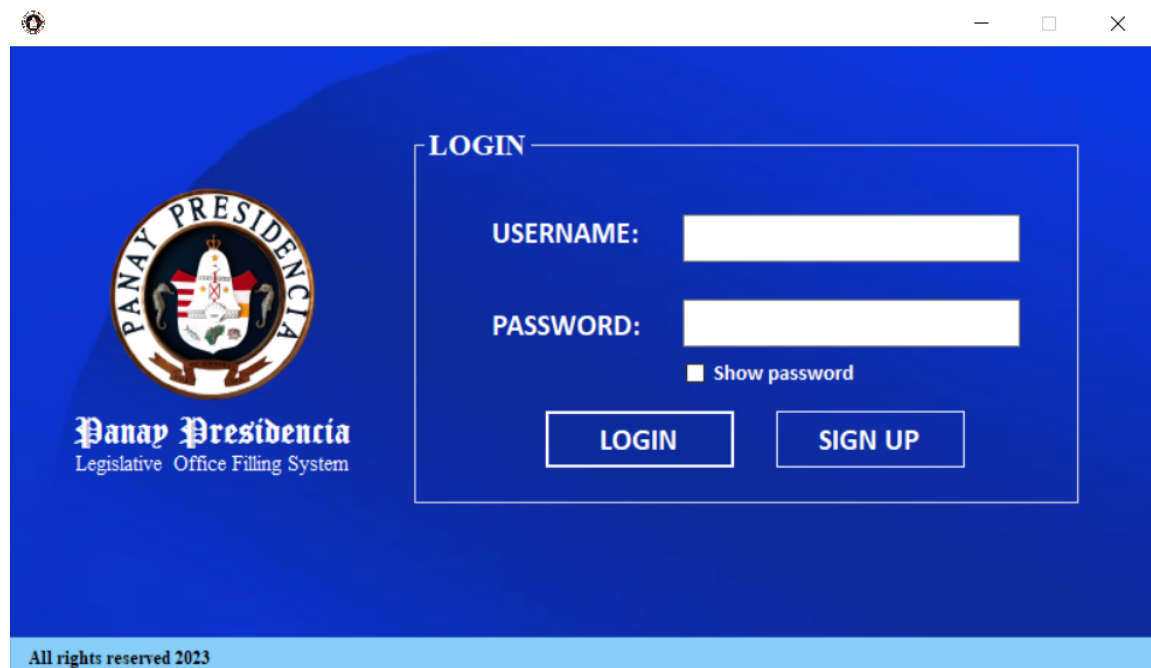
Print. If the user is the administrator, the admin can print documents and images from the system.

Database. This is where all documents and images are all stored and as well as user accounts and passwords are saved; it also corresponds to a local storage device where all the files will be saved.

The system

To the Municipality of Panay. The LGU of Panay, specifically the employees of the office of the Sangguniang Bayang, will be able to perform their tasks and services by viewing documents, more efficiently and all files and documents will be secured and backed up in the soft copy form.

The Vice Mayor's Office, specifically the Vice Mayor's Secretary, will use this system as an Admin Account, allowing them to conduct and access more broad functions. The administrator is responsible for managing the system, including user approval and file changes.



The image shows a web application window titled "Panay Presidencia Legislative Office Filing System". The background is blue with a large, faint watermark of the Philippine map. On the left, there is a circular seal of the Panay Presidencia and the text "Panay Presidencia Legislative Office Filing System". On the right, there is a "LOGIN" form with the following elements:

- USERNAME:** A text input field.
- PASSWORD:** A text input field with a "Show password" checkbox below it.
- LOGIN** and **SIGN UP** buttons.

At the bottom of the window, there is a light blue footer bar with the text "All rights reserved 2023".

Figure 5. Login Authentication of the system's users

The LOFSPanay login website will be available to the general public or to the office of the Sangguniang Bayan, with limited access to the system and will be restricted for security. The admin or the Office of the Vice Mayor will have better access and control of the system.



The image shows a registration form titled "REGISTRATION" in white text on a blue background. The form is divided into two sections by a horizontal line. The top section contains five input fields: "USERNAME:", "PASSWORD:", "RE-ENTER PASSWORD:", "DEPARTMENT:" (with a dropdown arrow), and "USER TYPE:" (with a dropdown arrow). The bottom section contains two input fields: "ADMIN USERNAME:" and "PASSWORD:". At the bottom of the form are two buttons: "SIGN UP" and "CANCEL".

Figure 6. Registration Form of the system's new users

The User Registration serves as the creating of an account for both the user and the admin user but would still need the permission personally to the Admin.



Figure 7. Homepage of the system

Upon entering the system through login authentication, the homepage will display all of the system's modules, including Ordinances, Resolutions, Committee Reports, Privilege Speeches, Journals, Internal Rules, Codes, Accounts (which will only show for the Admin Account), and About Us. Additionally, display the Panay Logo as the system's background.

OrdinanceNumber	OrdinanceDate	Sponsor	Committee	OrdinanceTitle	FilePath
100	2023-02-15	HON. JOHN	COMMITTEE ON FINANCE & ...	AN ORDINANCE APPROVING THE ANNUAL BUDGET OF THE MUNICIPALITY OF PANAY FOR FY 2023 IN THE AMOUNT OF ONE HUNDRED NINETY THREE MILLION NINE THOUSAND THREE HUNDRED SEVENTY NINE PESOS (P193,009,379.00) OF THE GENERAL FUND.	C:\Users\Client\Desktop\VISU...
33	2023-08-29	HON. ROGELIO A. BESORIO	COMMITTEE ON SOCIAL WEL...	55555555	C:\Users\Client\Desktop\VISU...
99	2023-08-29	HON. ALEXANDER RICHARD ...	COMMITTEE ON WOMEN, C...	aaaaaaaaaaaaaaaaaaaaaa	C:\Users\Client\Desktop\VISU...
33	2021-03-19	HON. ARLAN B. BAE	COMMITTEE ON INFRASTRU...	55222222222222222222	C:\Users\Client\Desktop\VISU...
8	2020-08-12	HON. ARLAN B. BAE	COMMITTEE ON AGRICULTU...	0000000000000000	C:\Users\Client\Desktop\VISU...

Figure 8. Ordinances Module

Ordinances Module contains all the ordinances files from the office, the user can only view and search, but an Admin user can not only just view but the admin can also edit, upload, delete, and print.

ResolutionNumber	ResolutionDate	ResolutionSponsor	ResolutionCommittee	ResolutionTitle	FilePath
1	2023-01-11	HON. RONILLO B. BADOLES	COMMITTEE ON FINANCE & ...	RESOLUTION EARNESTLY REQUESTING THE HONORABLE ARIEL F. NEPOMUCENO, UNDERSECRETARY, OFFICE OF THE CIVIL DEFENSE (OCD) AND EXECUTIVE DIRECTOR, NATIONAL DISASTER RISK REDUCTION AND MANAGEMENT COUNCIL (NDRMAC) FOR FINANCIAL ASSISTANCE IN THE AMOUNT OF ONE HUNDRED MILLION PESOS (P100,000,000.00) FOR THE CONSTRUCTION OF MULTI-PURPOSE EVACUATION CENTER IN POBLACION ILAVA, PANAY, CAPIZ	C:\Users\Client\Desktop\VISU...
3333	2023-08-29	HON. LUIS B. ASIS	COMMITTEE ON TRADE & IN...	888888888	C:\Users\Client\Desktop\VISU...
4	2023-08-29	HON. LUIS B. ASIS	COMMITTEE ON HEALTH AN...	789tyffgns	C:\Users\Client\Desktop\VISU...
99	2023-08-29	HON. ARLAN B. BAE	COMMITTEE ON INFRASTRU...	wwwwwwwwwwwwwww	C:\Users\Client\Desktop\VISU...

Figure 9. Resolutions Module

Resolutions Module contains all the resolution files from the office, the user can only view and search, but an Admin user can not only just view but the admin can also edit, upload, delete, and print.

CommitteeReportNumber	SubmittedByCommittee	Date	Subject	FilePath
1	w	2023-08-29	aaaaa	C:\Users\Client\Desktop\VISUAL STUD...
2	s	2023-08-29	aaaa	C:\Users\Client\Desktop\VISUAL STUD...
33	aaaaaaaaaaaa	2023-08-29	dddddddddd	C:\Users\Client\Desktop\VISUAL STUD...
99	ddddd	2022-11-24	dddddddddddddddddddddddddd	C:\Users\Client\Desktop\VISUAL STUD...

Figure 10. Committee Reports Module

Dashboard



Welcome
john

LOGOUT

Ordinances

Resolutions

Committee Reports

Privilege Speeches

Journals

Internal Rules

Codes

My Account

About Us


Privilege Speech

PrivilegeSpeechBy	Date	FilePath
a	2023-08-21	C:\Users\Client\Desktop\VISUAL STUDIO PROJECTS\APP\DM5pan...
f	2029-03-06	C:\Users\Client\Desktop\VISUAL STUDIO PROJECTS\APP\DM5pan...
oooooooooooo	2023-08-29	C:\Users\Client\Desktop\VISUAL STUDIO PROJECTS\APP\DM5pan...

Dashboard



Welcome
john

LOGOUT

- Ordinances
- Resolutions
- Committee Reports
- Privilege Speeches
- Journals**
- Internal Rules
- Codes
- My Account
- About Us


Journals

Date	FilePath
2023-08-03	C:\Users\Client\Desktop\VISUAL STUDIO PROJECTS\APP\DMspanay (8-29-23)\FILES\JOURNALS\1_wdwpaeodoeedleded_patrick_04_23_2001_health.docx
2023-08-29	C:\Users\Client\Desktop\VISUAL STUDIO PROJECTS\APP\DMspanay (8-29-23)\FILES\JOURNALS\2023\ERLE LETTER.docx
1991-04-25	C:\Users\Client\Desktop\VISUAL STUDIO PROJECTS\APP\DMspanay (8-29-23)\FILES\JOURNALS\1991\SECRETARY'S CERTIFICATE.docx

Journals Module contains all the Journals from the office, the user can only view and search, but an Admin user can not only just view but the admin can also edit, upload, delete, and print.

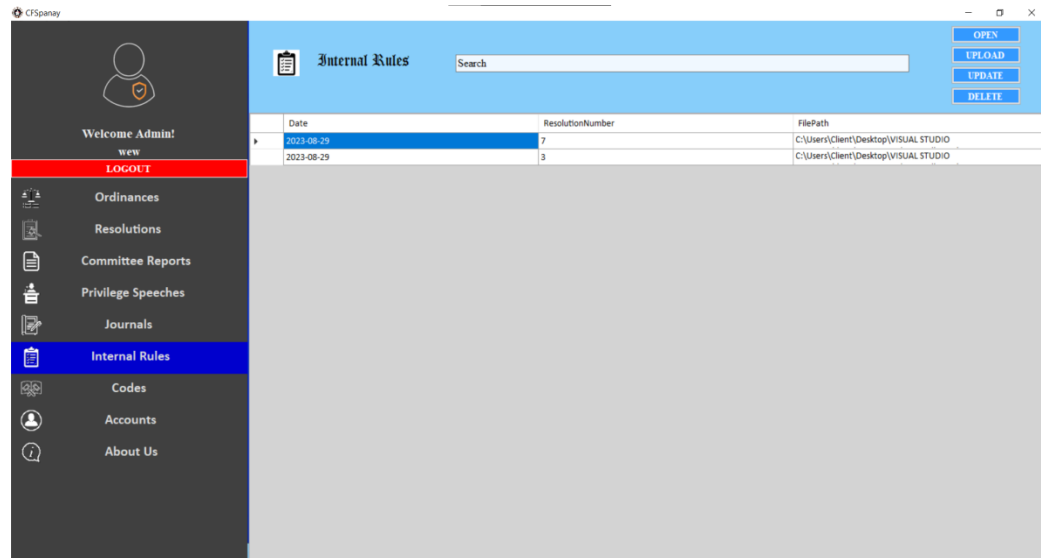


Figure 13. Internal Rules Module

Internal Rules Module contains all the Internal Rules document from the office, the user can only view and search, but an Admin user can not only just view but the admin can also edit, upload, delete, and print.

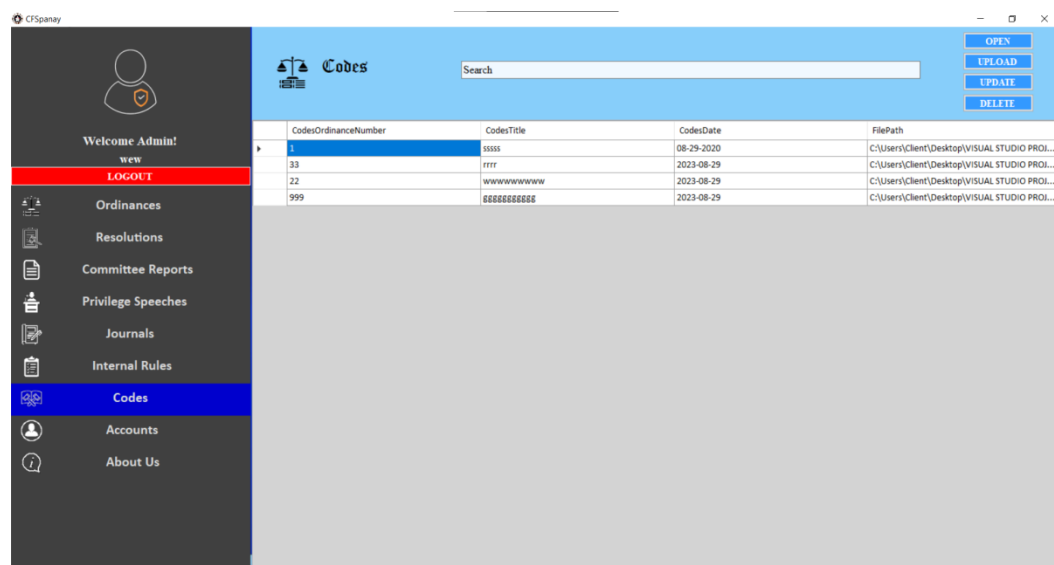


Figure 14. Codes

Codes Module or Codification all permanent ordinances adopted by the governing body into a code book, the user can only view and search, but an Admin user can not only just view but the admin can also edit, upload, delete, and print.

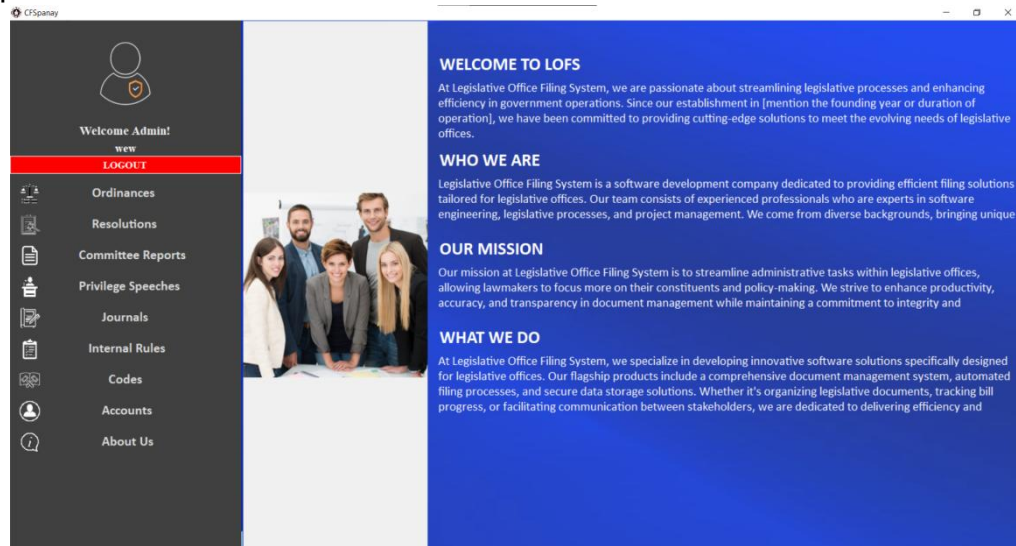


Figure 15. About Us Module

About Us Module is all about the system and both the User and the Admin Account can only view in this module.

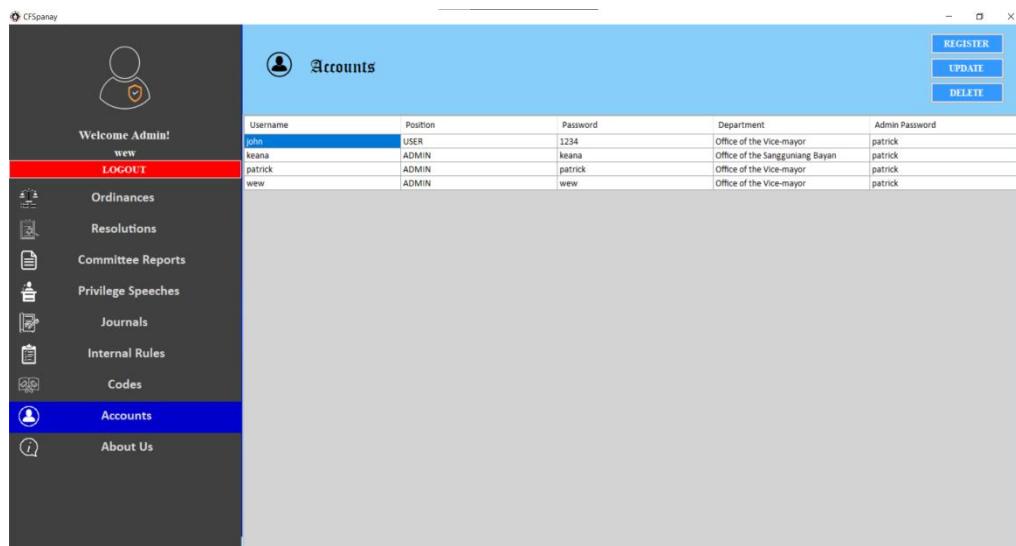


Figure 16. Accounts Module

The Accounts Module contains all of the office's registered users, including Sangguniang Bayand and the vice-mayor. This module is only accessible to the Admin Account because it can read, edit, and remove all of the user's information, as well as register or add new users using the same authentication method as the log in.

A screenshot of a web-based registration form titled "REGISTRATION" in white text on a blue background. The form is enclosed in a white border. It contains several input fields: "USERNAME:" with a text box, "PASSWORD:" with a text box, "RE-ENTER PASSWORD:" with a text box, "DEPARTMENT:" with a dropdown menu, and "USER TYPE:" with a dropdown menu. Below these fields is a horizontal separator line. Under the line, there are two more input fields: "ADMIN USERNAME:" and "PASSWORD:". At the bottom of the form are two buttons: "SIGN UP" and "CANCEL".

Figure 17. Register under Accounts Module

In Accounts Module, admin can add new users and in this section an admin can input details from that user.

Dashboard

MY ACCOUNT

USERNAME: John

OLD PASSWORD:

ENTER NEW PASSWORD:

RE-ENTER NEW PASSWORD:

UPDATE CANCEL

Dashboard

Welcome John

LOGOUT

Ordinances

Resolutions

Committee Reports

Privilege Speeches

Journals

Internal Rules

Codes

My Account

About Us

Figure 18. My Account

In My Account Module, the user can update their old password into a new password, with confirmation of the old one.

Physical environment

Hardware Specification

Client

CPU Type Model	IntelCore i7
Memory	8.00 GB Ram
Storage	500 SSD storage
Input Devices	Mouse and Keyboard
Output Device	Monitor

Software Specification

Front End	Visual Studio
Back End	C# and MySQL

Front End: Visual Studio

Visual Studio is an Integrated Development Environment(IDE) developed by Microsoft to develop Desktop applications, GUI(Graphical User Interface), console, web applications, mobile applications, cloud, and web services, etc. With the help of this IDE, you can create managed code as well as native code. It uses the various platforms of Microsoft software development software like Windows store, Microsoft Silverlight, and Windows API, etc. It is not a language-specific IDE as you can use this to write code in C#, C++, VB(Visual Basic), Python, JavaScript, and many more languages. It provides support for 36 different programming languages. It is available for Windows as well as for macOS.

Visual Studio IDE is a full-featured programming environment that supports several operating systems, the web, and the cloud. Users can easily navigate the UI, allowing them to write their code quickly and precisely.

Visual Studio includes a powerful debugger to assist developers in quickly identifying any code issues. (geekforgeeks.org, 2013)

Back End: C# Language

C# is a modern, innovative, open-source, cross-platform object-oriented programming language and one of the top 5 programming languages on GitHub. C# is the most popular language for .NET development. With .NET you can target any application type running on any platform. Reuse your skills, code, and favorite libraries across all of them in a familiar environment. That means you can build apps faster, with less cost. From mobile applications running on iOS and Android, to enterprise server applications running on Windows Server and Linux, or high-scale microservices running in the cloud, .NET provides a solution for you. (dotnet.microsoft, 2024)

RDBMS: MySQL

MySQL is the most widely used open-source database. According to DB-Engines, MySQL is the second-most popular database, trailing Oracle Database. MySQL underpins many of the most popular applications, such as Facebook, Twitter, Netflix, Uber, Airbnb, Shopify, and Booking.com.

MySQL is open source, thus it includes many features that have been developed in close collaboration with users over the past 25 years. MySQL Database most likely supports your preferred application or programming language.(Oracle, 2024)

Cost of development

Hardware Development Cost

Web Server System Unit	Php 60, 000.00
Keyboard and Mouse	Php 750.00
Printer	Php 10,000.00
UPS	Php 8,500.00
Total Cost	Php 79,250.00

System Development Cost

System Analyst	Php 25,000.00
Programmer	Php 40,000.00
GUI Designer	Php 15,000.00
Researcher	Php 15,000.00
Total Cost	Php 95,000.00

System Depreciation Cost

Hardware depreciation cost

$$\text{Annual Hardware Depreciation} = \frac{\text{Php 79,250.00}}{5 \text{ years}}$$

$$\text{Annual Hardware Depreciation Value} = \text{Php 15,850.00}$$

Year	Total Balance
0	Php 79,250.00
1	Php 63,400.00
2	Php 47,550.00
3	Php 31,700.00
4	Php 15,850.00
5	Php 0.00

Software Depreciation

$$\text{Annual Software Depreciation} = \frac{\text{Php } 95,000.00}{10 \text{ years}}$$

$$\text{Annual Software Depreciation} = \text{Php } 9,500.00$$

Year	Total Balance
0	Php 95,000.00
1	Php 85,500.00
2	Php 76,000.00
3	Php 66,500.00
4	Php 57,000.00
5	Php 47,500.00
6	Php 38,000.00
7	Php 28,500.00
8	Php 19,000.00
9	Php 9,500.00
10	Php 0.00

Tangible and Intangible Benefit

Tangible Benefit

- Saves time by making it easier to manage and find documents.
- Protects sensitive information with user authentication and access controls.
- Safeguards documents against loss or damage with automatic backups.
- Reduces expenses on physical storage and document printing.

Intangible Benefits

- Improves workplace morale with user-friendly tools and streamlined processes.
- Allows officials to make informed decisions by providing accurate and up-to-date information.
- Reduces paper waste and supports sustainability efforts.
- Despite initial costs, the system can result in long-term cost savings through increased efficiency and decreased spending.

Architectural Design

Use Case Diagram of the System

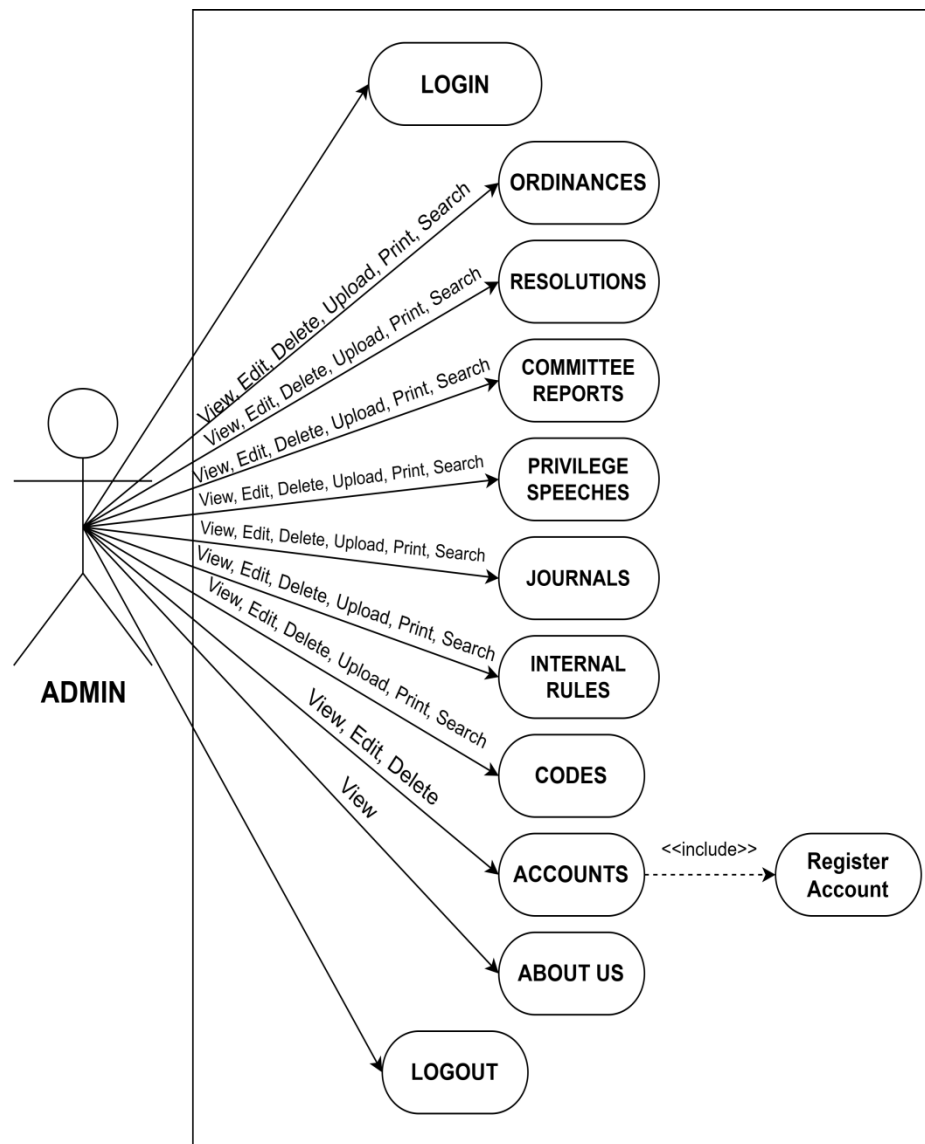


Figure 19. Use Case Diagram of the System - Admin

The use case diagram presents the actions that users depict the processes that will be used. For the admin user the system has the same tasks for several modules such as Ordinances, Resolutions, Committee Reports, Privilege Speeches, Journals, Internal Rules, and Codes by viewing, updating, deleting, upload and print important documents.

However, for the About Us module you can only view as it only describes the system. Lastly, for the Accounts module it generally modifies all user accounts and this module will only be for the Admin account.

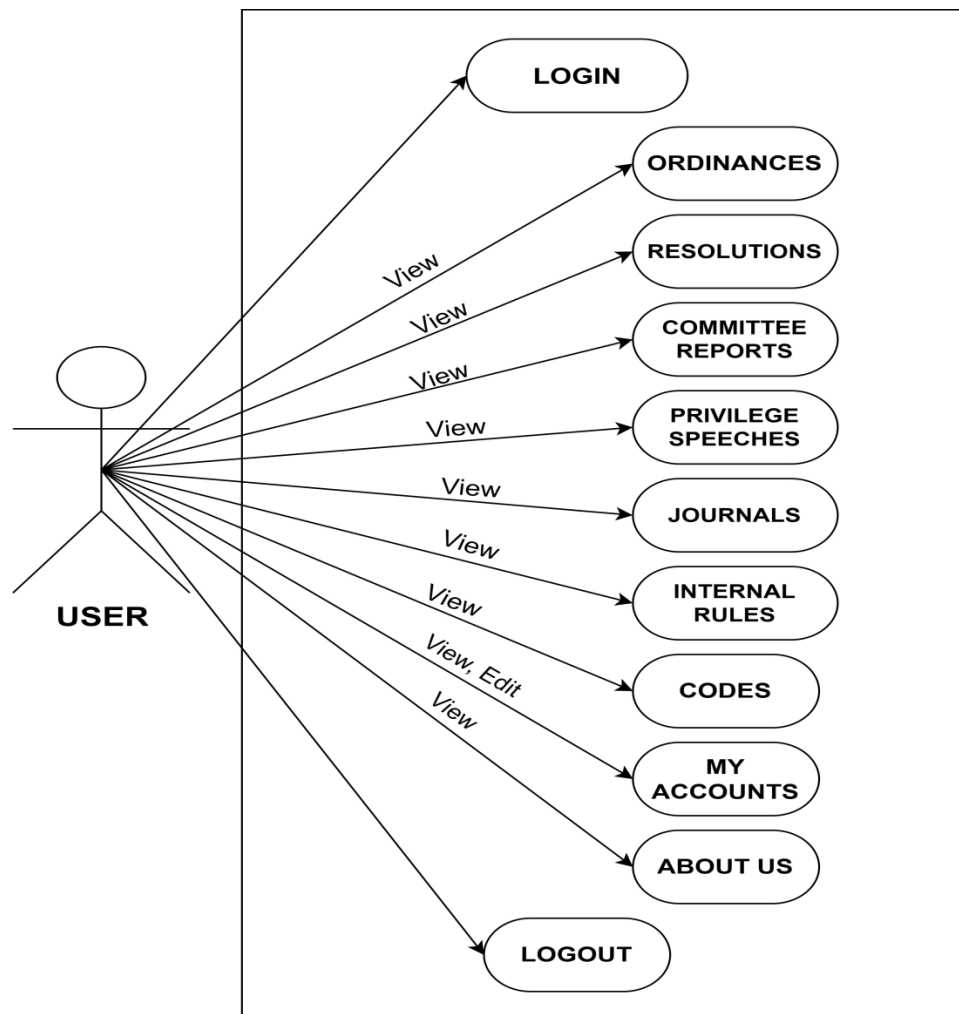


Figure 20. Use Case Diagram of the System - User

The use case diagram depicts the actions that users take and the processes that will be used. The user account or the Sangguniang Bayan office employee is limited to viewing each module. The Account Module is only available to the Administrator for security reasons, hence the User account does not have one.

Sequence diagram of the system

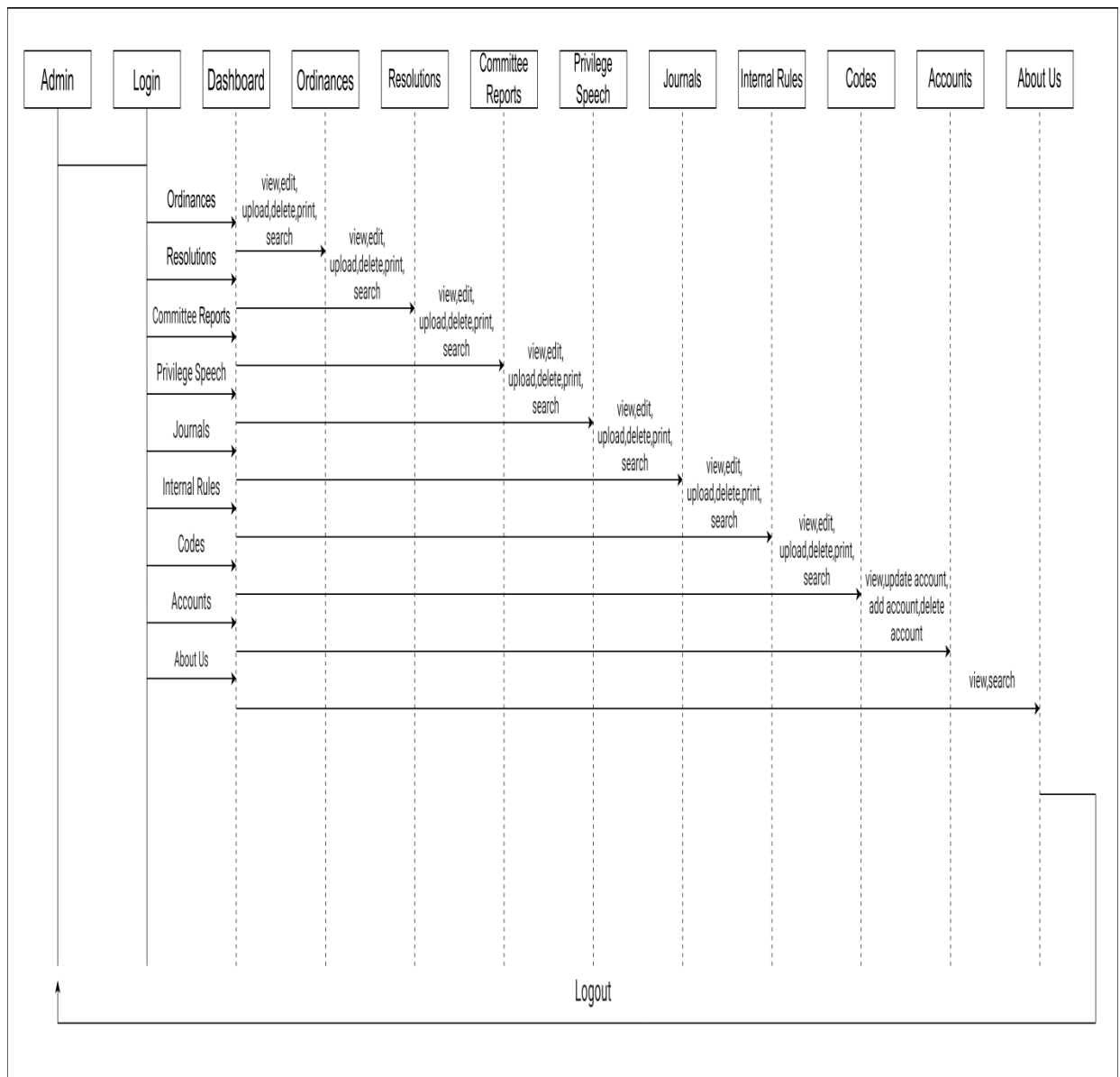


Figure 21. Sequence Diagram - Admin

The sequence diagram of the system for the administrator, as presented in Figure 21 shows the detailed actions where in the start of the processes initiates on the Log-in. Upon logging in to the system it will automatically direct you to the dashboard which includes modules such as Ordinances, Resolutions, Committee Reports, Privilege Speeches, Journals, Internal Rules, and Codes

where it has the same function that an Admin can view, update, upload, delete and print. For the About us module the admin can only view. The main function of the Admin is to manage and modify Accounts, the admin can view, update, add, and delete an Account.

Activity / Diagrams of the system

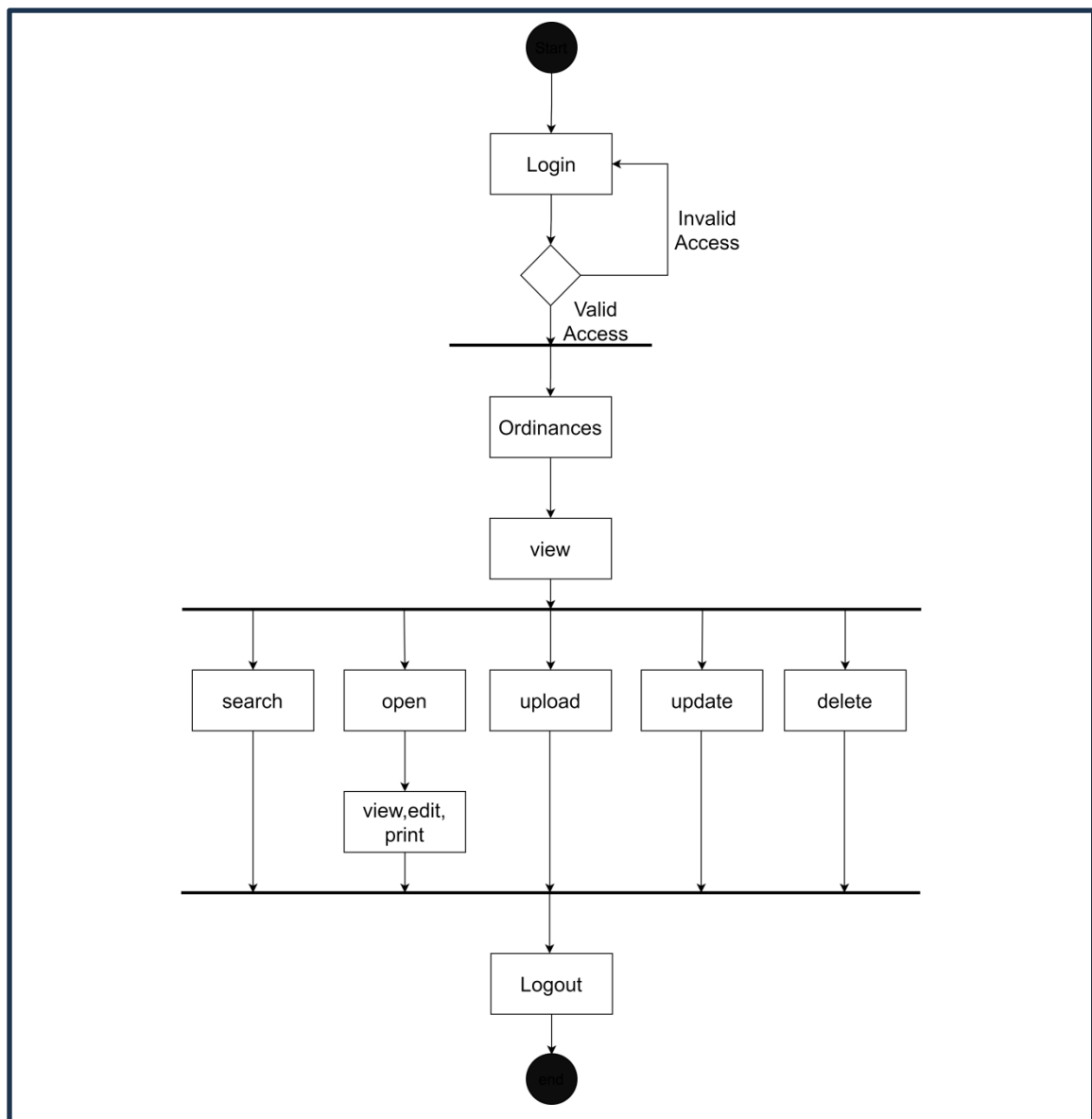


Figure 22. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Ordinances module is selected, the administrator can view, edit, print, delete, search, and add documents. However, User can only view its content.

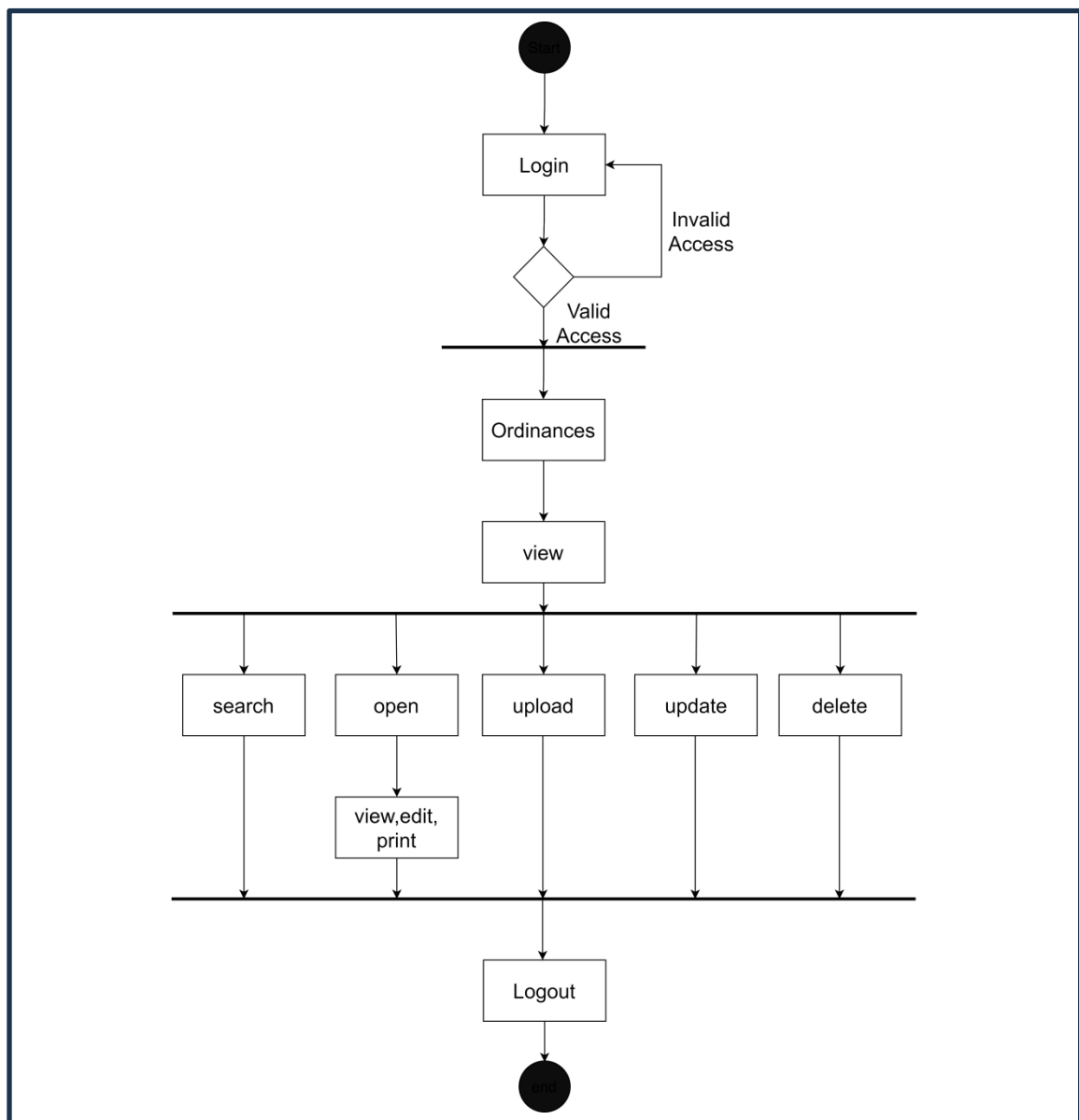


Figure 23. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Resolutions module is selected, the administrator can view, edit, print, delete, search, and add documents. However, User can only view its content.

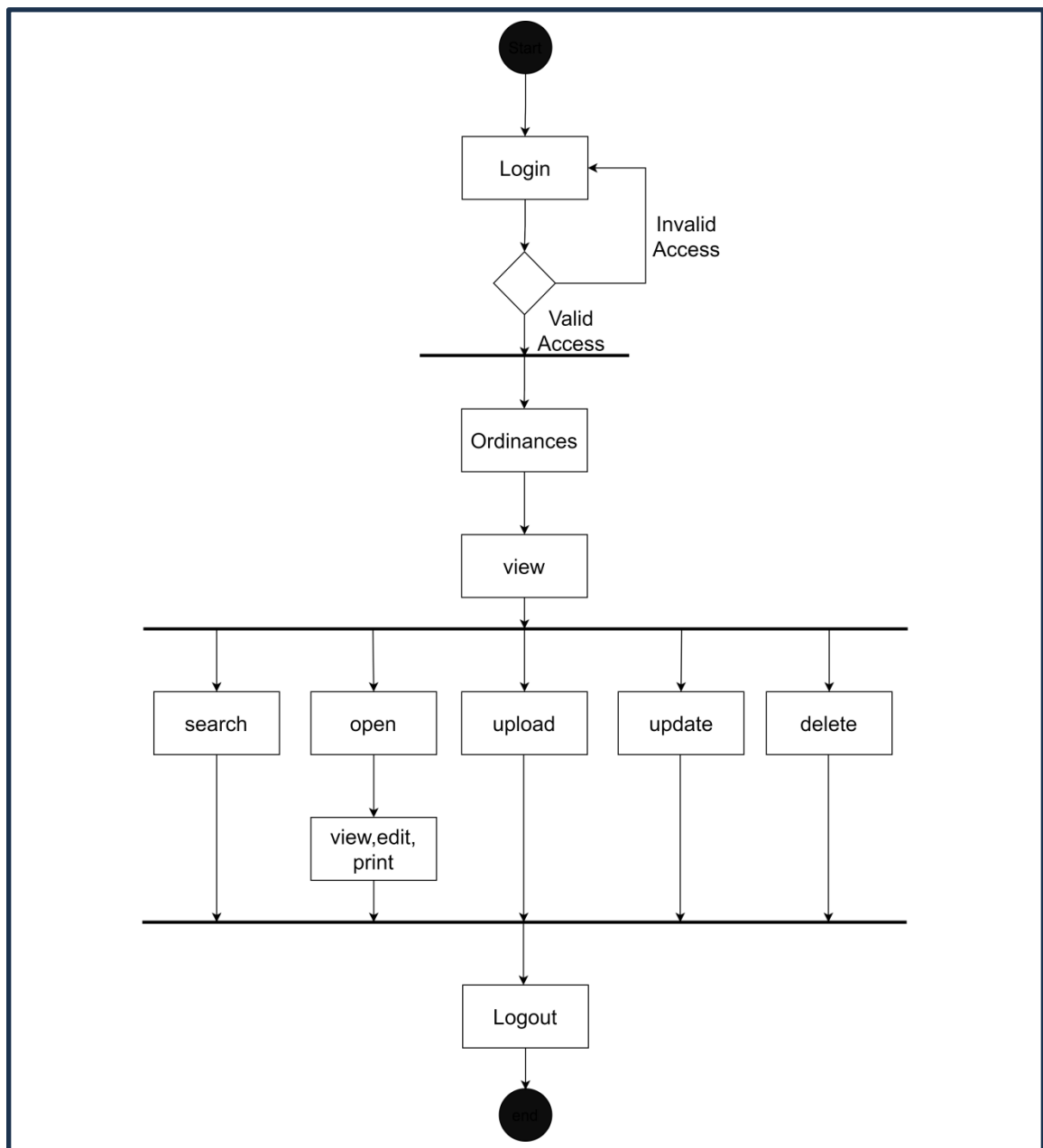


Figure 24. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Committee Reports module is selected, the administrator can view, edit, print, delete, search, and add documents. However, User can only view its content.

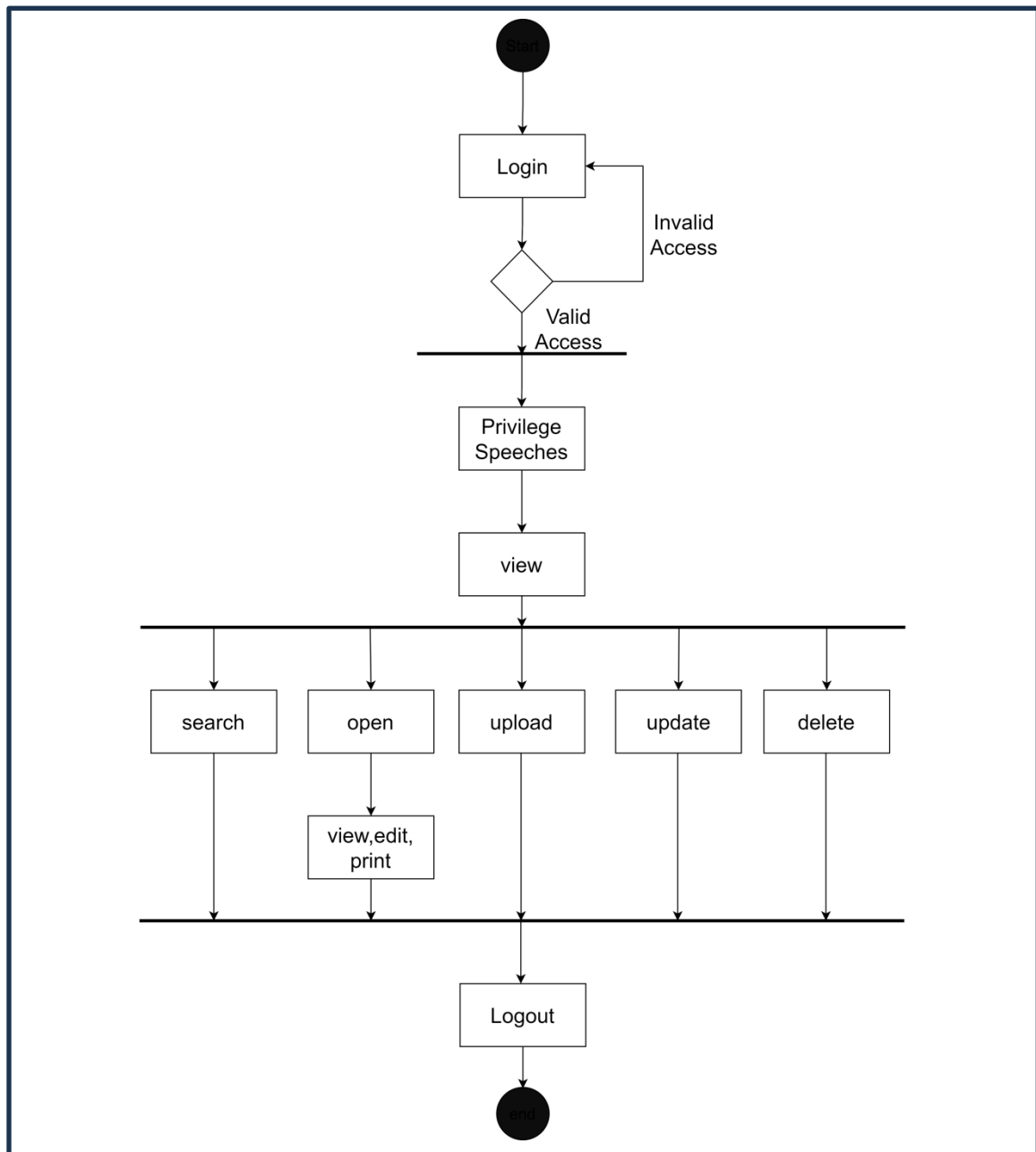


Figure 25. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Privilege Speeches module is selected, the administrator can view, edit, print, delete, search, and add documents. However, User can only view its content.

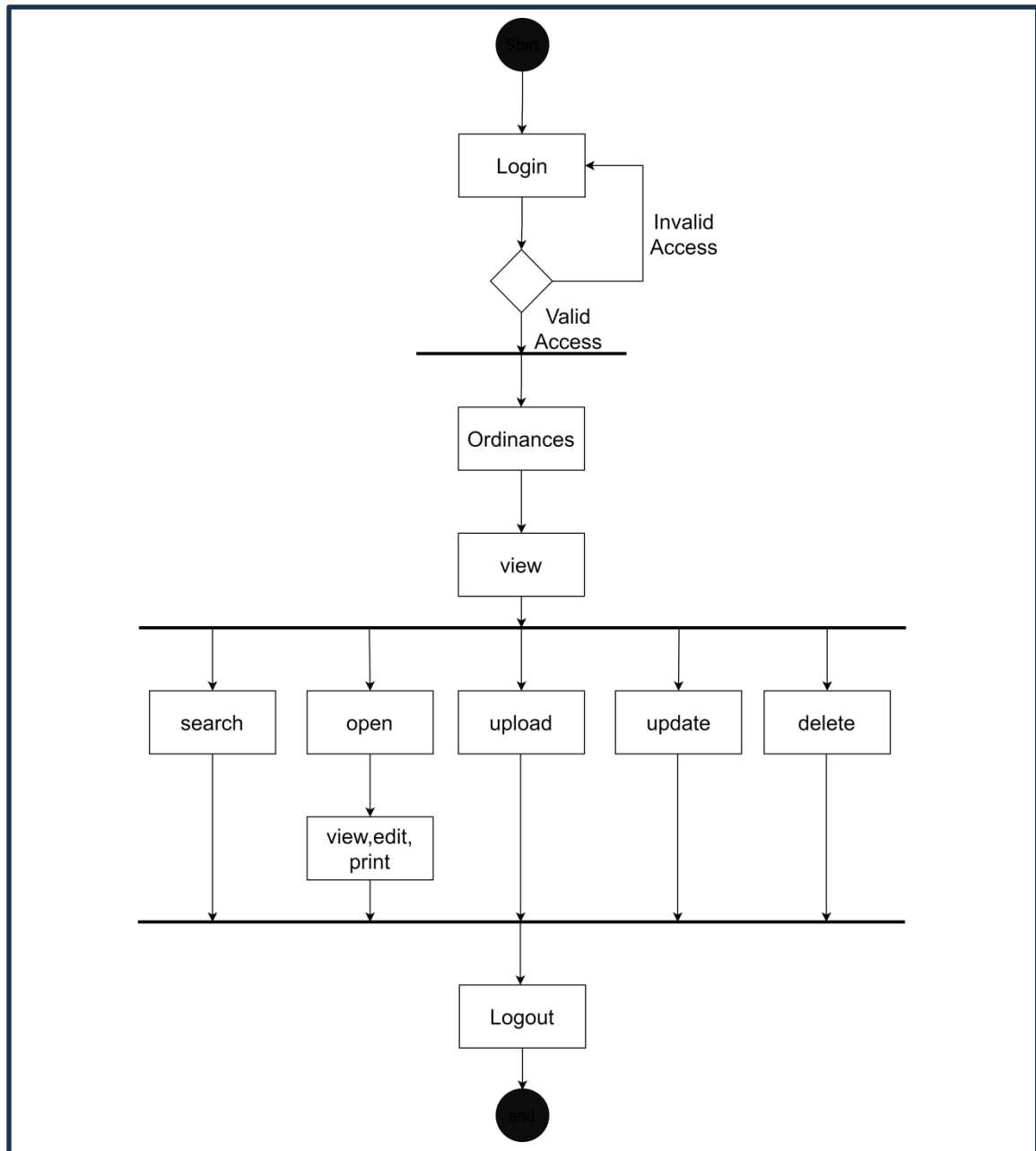


Figure 26. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Journals module is selected, the administrator can view, edit, print, delete, search, and add documents. However, User can only view its content.

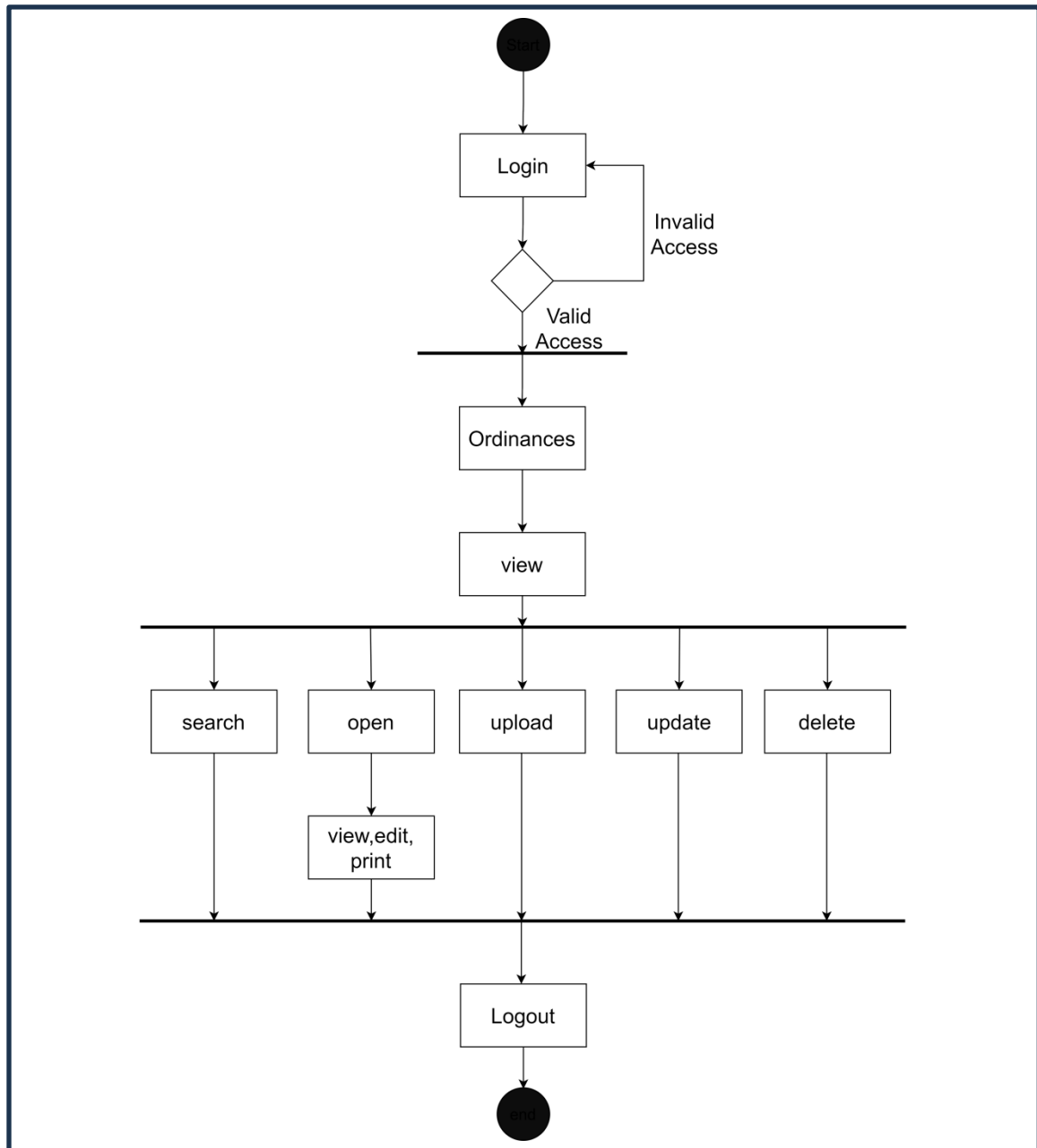


Figure 27. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Internal Rules module is selected, the administrator can view, edit, print, delete, search, and add documents. However, User can only view its content.

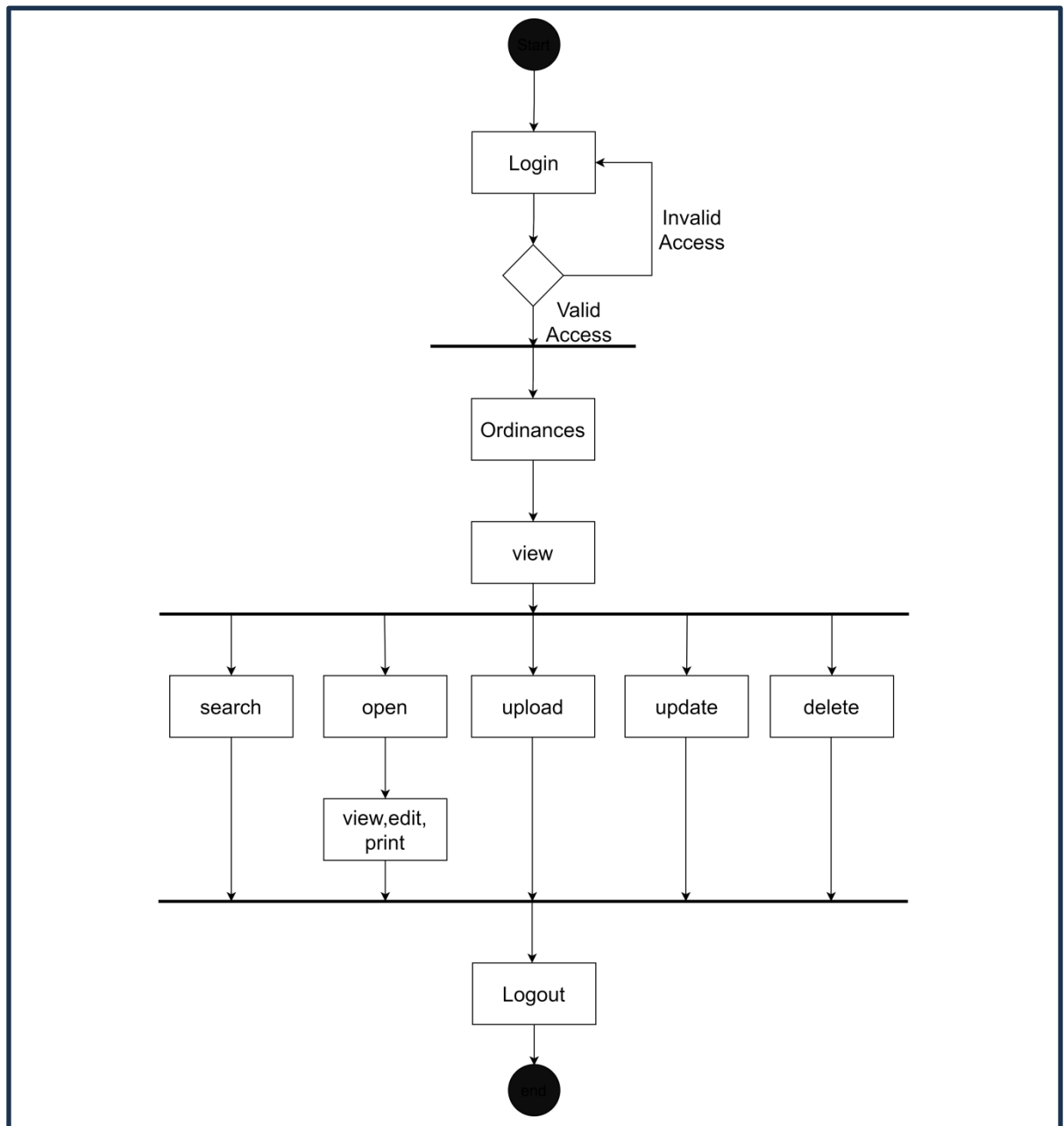


Figure 28. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Codes module is selected, the administrator can view, edit, print, delete, search, and add documents. However, User can only view its content.

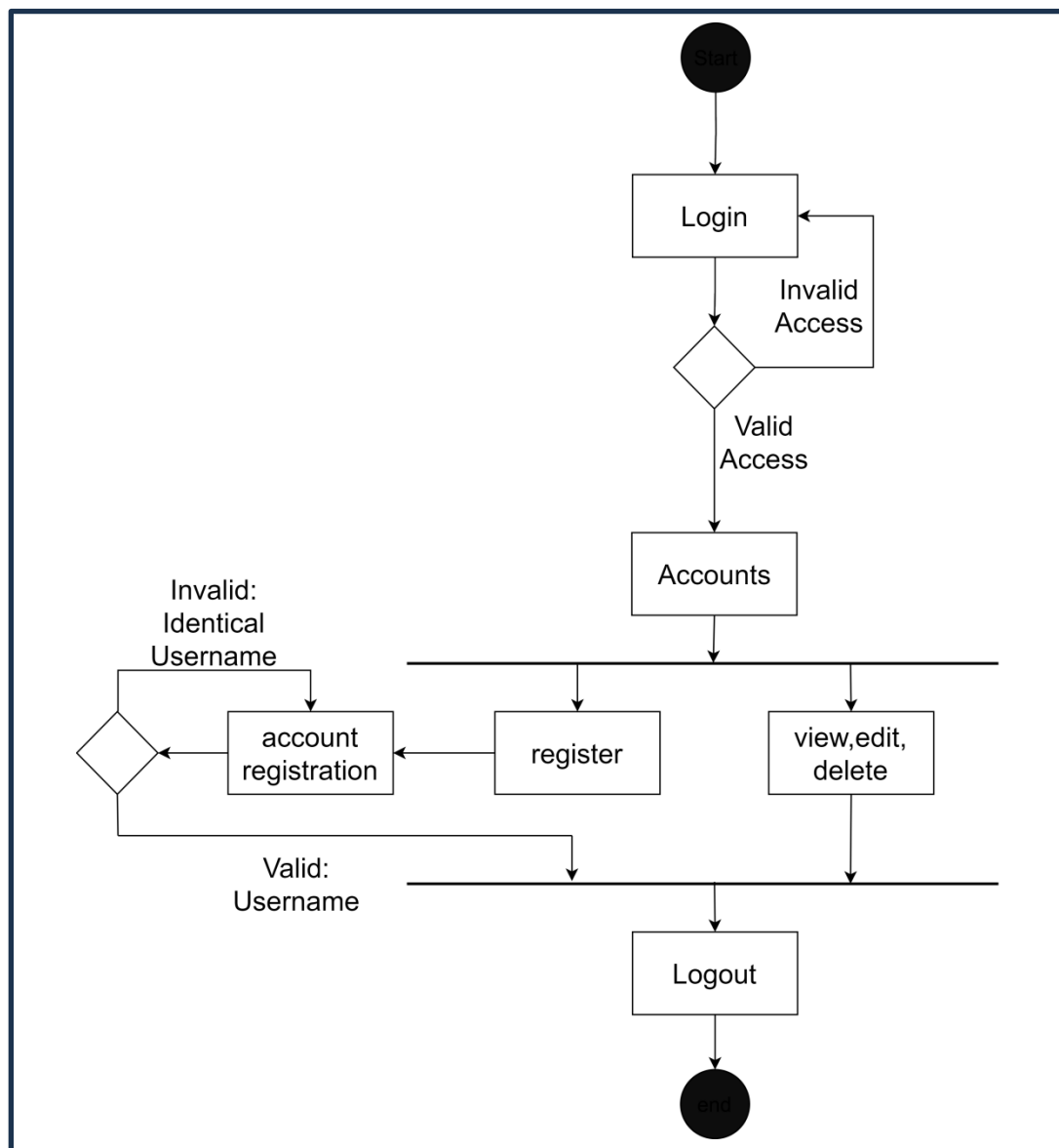


Figure 29. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the Accounts module is selected, the administrator can view personal details, edit, add or register and delete an account. However, user cannot access or cannot be seen inside the user interface.

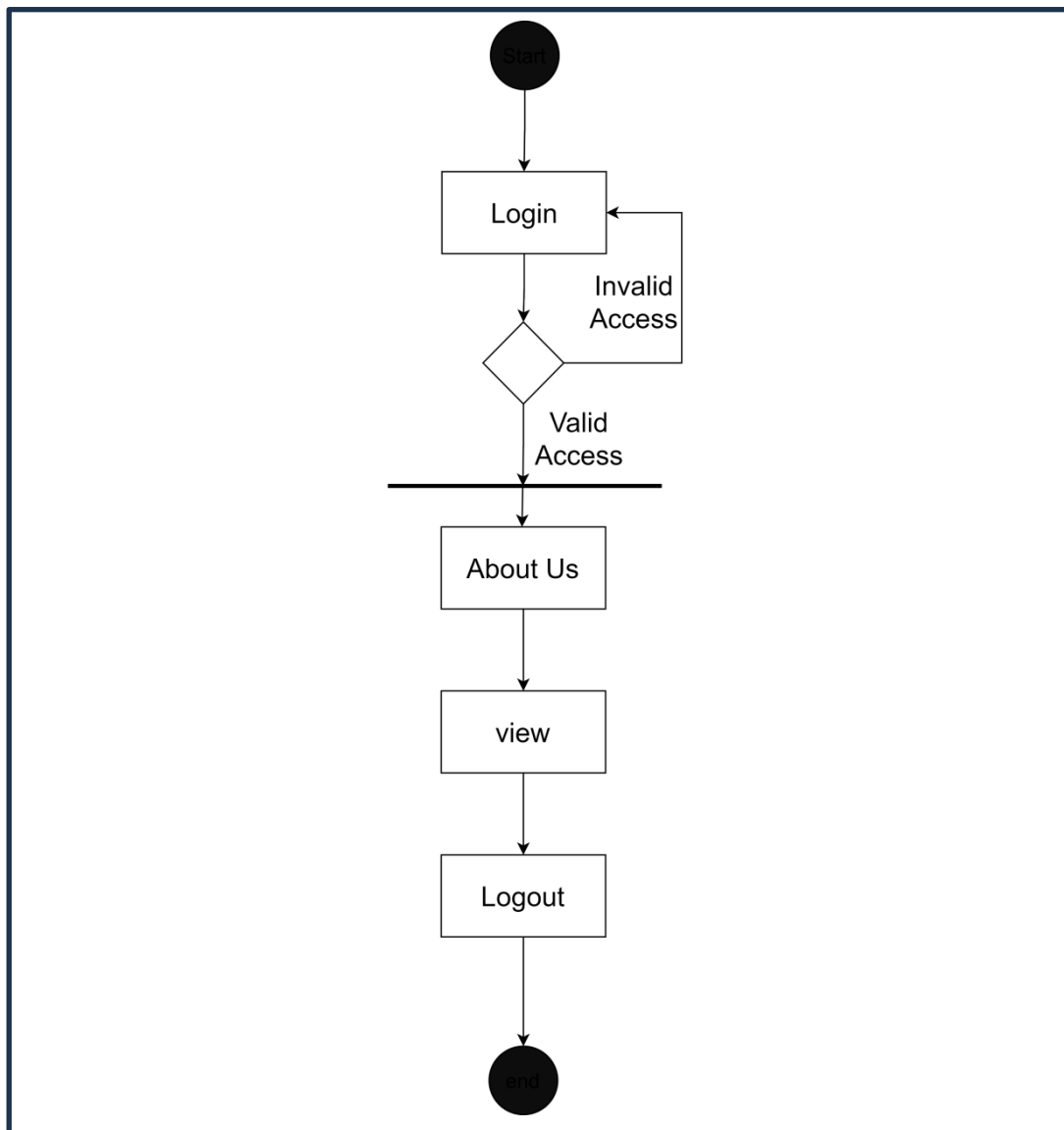


Figure 30. Activity Diagram of the System

The system's activity diagram shows the entire workflow of tasks that the administrator can do. Each process is a step-by-step guide to system access. Once the administrator has signed in, if the About Us module is selected, the administrator and user can both view the module's content.

Design and Implementation Issues

In the development process of the system, a deep understanding of the functions and operations of the Sangguniang Bayan and the office of the Vice Mayor of Panay was imperative. This comprehension was acquired through comprehensive interviews conducted directly with the personnel of these offices. These interactions informed the decision-making process of the system developer, leading to the creation of a system tailored to address a significant challenge faced by these offices.

The system's interface was meticulously crafted to align with the preferences and requirements of the client. In the context of LOFSPanay, the system primarily operates on personal computers (PCs), with provisions in place to seamlessly adapt to various screen resolutions across different devices.

Designing the graphical user interface posed notable challenges during the development phase. However, diligent efforts were invested to refine and enhance the interface, ultimately culminating in the creation of a user-friendly environment accessible to all users of the system.

Moreover, adjustments were made to text labels and report formatting to ensure coherence and clarity, particularly when generating printed reports from the collected data. These modifications aimed to enhance the comprehensibility and professional presentation of the reports derived from the system's data.